FictiPay Customer Churn Prediction

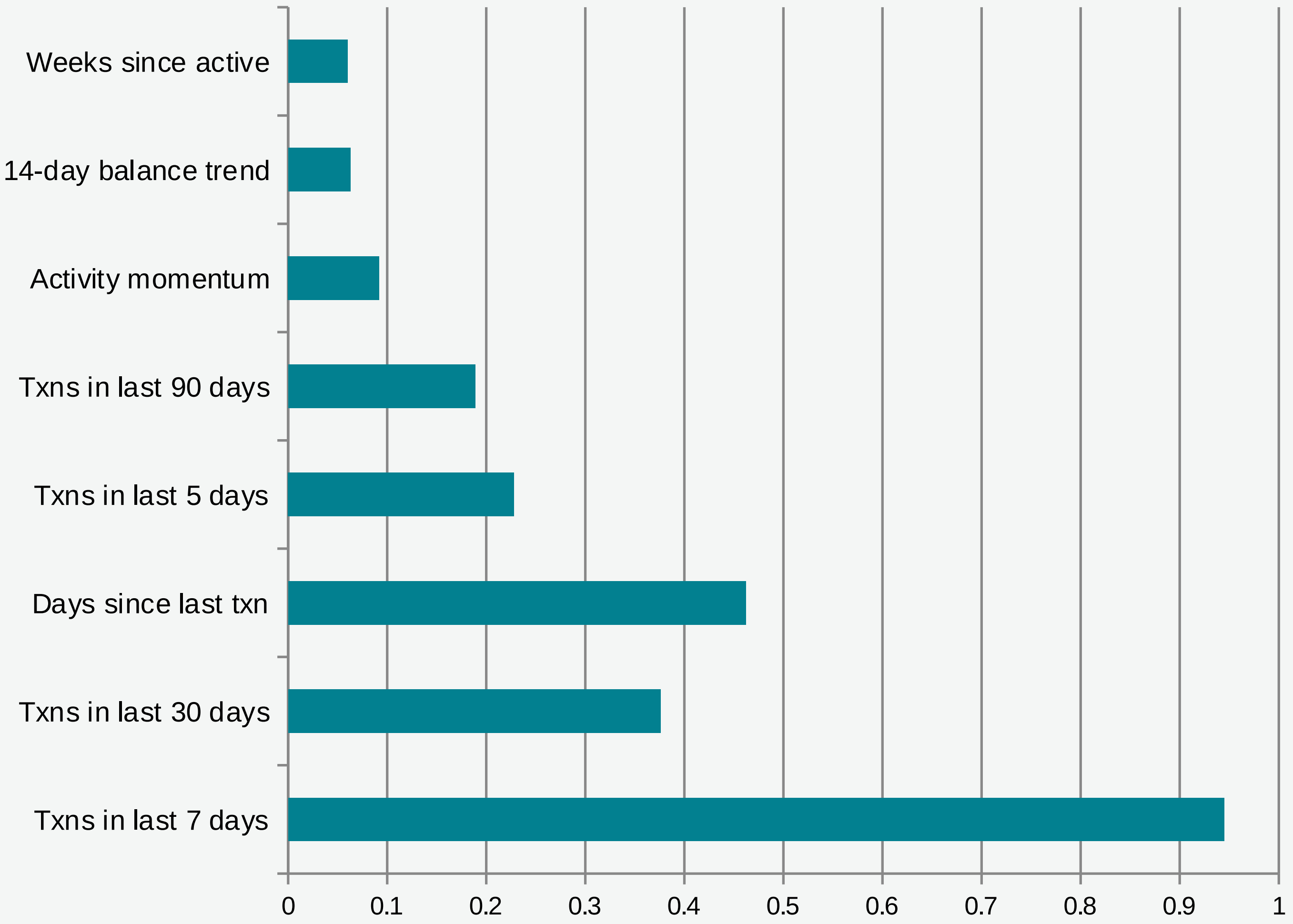
Identifying at-risk customers from transaction and balance behavior

850,000	200M+	137	0.9846
Customer accounts scored	Transaction records (Jan–Mar 2024)	Engineered features	Final OOF AUC-ROC

Approach: Dask-based feature engineering over the full transaction and balance history (recency, dormancy streaks, balance trend, channel mix) feeding a LightGBM + CatBoost ensemble, evaluated with 10-fold × 3-seed cross-validation.

What Drives Churn

Top model signals, ranked by impact on the predicted churn probability (SHAP)



Recent activity level

Customers whose transaction count over the last 7–30 days has dropped sharply are by far the strongest churn signal.

Recency of last transaction

Time since the customer's most recent transaction of any type is the second-strongest signal.

Balance trajectory

A negative 14-day balance trend and a long run of weeks with no activity add secondary, confirming risk.

A Practical Decision Rule

Flag the top 10% of accounts by predicted churn probability for retention outreach

91.0%	71.8%	0.985
Precision @ Top 10%	**Recall @ Top 10%**	**AUC-ROC (OOF)**
9 in 10 flagged accounts genuinely churn	Captures almost three-quarters of all churners	Overall ranking quality of churn risk scores

Reading the numbers

Out of every 100,000 accounts, the top-scored 10,000 contain roughly **9,100 true churners (precision)** and represent **72% of all churners in the population (recall)**. This concentrates retention spend on a small, high-confidence list rather than spreading it across the whole customer base.

Recommended Interventions

Mapping each risk signal to a concrete retention action

Risk Signal	What it Means	Recommended Action
Sharp drop in 7–30 day transaction count	Customer has gone quiet very recently — the strongest early-warning signal	Personalized re-engagement push notification with a small cashback or fee-waiver offer on next transaction
Long stretch since last transaction	Account has been dormant for an extended period across all channels	Time-limited reactivation campaign (bonus on next CashIn / P2P)
Negative 14-day balance trend	Balance is being drawn down, often preceding full disengagement	Proactive incentive (savings nudge, cashback) tied to maintaining balance
Missed expected bill payment	A previously-recurring bill payment did not occur this month	Reminder notification + prompt to enroll in auto-pay

Expected Impact & Next Steps

Expected Impact

- Targeting the top-10% risk list (91% precision) concentrates retention budget where it converts best, instead of spreading offers across the full 850K-account base.
- Recovering even a modest share of the ~72% of churners captured at this tier directly offsets the cost of the retention offers issued to the smaller flagged group.
- Channel-specific signals (e.g. missed bill payments, dormant P2P) allow offers to be tailored per customer rather than generic.

Limitations & Next Steps

- Built and validated on a synthetic dataset; real-world performance should be re-validated on production data.
- OOF AUC has plateaued around 0.985 across several feature-engineering iterations — likely near the ceiling for this aggregate-feature representation.
- Remaining hard cases: active accounts with rising balances that still churn, and quiet accounts with falling balances that don't — candidates for “last transaction anomaly” features.
- Next: sequence-based models over raw transaction order, and threshold/calibration tuning for production deployment.